galactose and mannose as well as Ketoses or sugars containing ketone group(e.g.) Fructose.

Major monosaccharides

a) **Glucose** (Dextrose or grape sugar): It serves as the main source of energy in the body. It is abundantly found in nature. It is found in sweet fruits such as grapes, berries, oranges in vegetables like sweet corn and carrots. It is less sweet than cane sugar. It is the end product in the digestion of disaccharides and polysaccharides and is the form of carbohydrate circulating in the blood.

b) **Fructose**(Levulose or fruit sugar): It is much sweeter than cane sugar and is found in honey,ripe fruits and some vegetables.It is also a product of the hydrolysis of sucrose.

c) **Galactose:** It does not occur in the free state, but occurs as a constituent of lactose present in milk.

B) Disaccharides

They are formed by the combination of 2 monosaccharides. The disaccharides of nutritional importance are sucrose, maltose and lactose.

a) **Sucrose** (Cane sugar, beetsugar, table sugar): It occurs in sugarcane (10-12%) and beetroot(12-18%).In the intestine, sucrose is broken down into monosaccharides -glucose and fructose by the enzyme sucrase present in the intestinal juice and then absorbed.

Sucrose —(Hydrolysis)→ **Glucose+fructose**

b) **Lactose**(Milk sugar): It occurs in the milk of mammals. Cow's milk and buffalo's milk contain 4% of lactose, while human milk contains about 7% of lactose. Lactose is hydrolysed to glucose and galactose by the enzyme lactase present in the intestinal juice.

Lactose —(Hydrolysis)→ **Glucose+galactose**

c) **Maltose** (Malt sugar): It is found in all sprouted and malted products.It is an intermediate product formed in the process of conversion of starch into glucose. Maltose is hydrolysed to 2 molecules of glucose by the enzyme maltase present in the intestinal juice. Sprouted cereals and beer contain large amount of maltose.

Maltose —(Hydrolysis)→ **Glucose+glucose**

9.2.2 Complex Carbohydrates

These are complex compounds with high molecular weights. Their structural formula is $(C_6H_{10}O_5)n$, where n>2.They are formed by a combination of more than 2 molecules of a monosaccharide. Unlike the sugars,which contain 3 monosaccharides – Glucose, fructose and galactose in different combinations, the polysaccharides – Starch and Glycogen are composed entirely of glucose. They differ from each other only in the nature of the bonds that link the glucose units together.

1) **Starch :** It is a long, straight or branched chain of hundreds of glucose units linked together. The important sources of starch are

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